Tower-Index Acceptance Map Integration:
Data-Driven Dead-Tower Masks + Cross-Section Systematic

PPG12 Analysis Team (automated report)

2026-04-22

Executive Summary

Four-level tower-index acceptance maps ($h_{\text{etaphi\_tower\_\{preselect,common,tight,tight\_iso\}\_\{mc\_inclusive,data\}}}$) are now populated automatically in the official efficiency pipeline and written to `Photon_final_bdt_nom.root` for every production. Direct data-vs-MC comparison shows clear dead-region asymmetry: MC does not emulate the full set of dead towers active in Run-24 data.

Primary methodology: data-only phi-symmetry. For each 2-eta-row band, fit a Gaussian to the 256 per-phi data counts and flag phi bins with z < -2 as dead. Independent of MC; exact under the assumption that the cross-section is phi-symmetric at fixed η in pp. Three masks generated — one per selection level: `mask_phisymm_{preselect, common, tight, or}` with 1344 / 1160 / 646 / 1180 masked towers respectively (the OR mask is the union of the common and tight masks).

Direct cross-section measurement: re-run the efficiency + unfolding + ABCD pipeline with the phi-symm masks applied symmetrically to both data and MC. Integrated σ shifts in $E_T^\gamma \in [10, 36]$ GeV:

variant	$\Delta\sigma/\sigma_{\text{nom}}$
`mask_phisymm_preselect`	$+2.01\%$
`mask_phisymm_common`	$+2.26\%$
`mask_phisymm_tight`	$+1.37\%$
`mask_phisymm_or` (common $\cup$ tight)	$+2.28\%$

Recommended envelope: **+2.5% one-sided acceptance systematic** on the nominal all-range cross-section. The four variants agree within $\sim$1% across selection levels, consistent with a uniform detector-level dead-tower effect.

Nominal: all-z (beam-delivered) lumi $64.37\,\text{pb}^{-1}$ + $v_{z,\text{truth}} < 9999$ (since 2026-04-22). Integrated cross-section $\sigma_{\text{nom}} = 1963\,\text{pb}$ over $E_T^\gamma \in [10, 36]$ GeV.

1 Findings Table

Severity	Location	Finding	Impact
HIGH-1	`Photon_final_bdt_nom.root`	MC GEANT acceptance does not emulate the full Run-24 data dead-tower list. Direct σ shift under symmetric masking is +1.4 to +2.3%.	Cross-section biased LOW by $\sim$2%; supply this as systematic or fix the MC dead-tower list.

INFO-1	`RecoEffCalculator_TTreeReader.C`	4 tower-index TH2F maps now auto-filled every production. Three optional mask config keys: `tower_mask_on/file/name`. Veto applied symmetrically to data + MC at top of cluster loop.	Integration complete. Masks become a drop-in variant.
INFO-2	`tower_masks_bdt_nom.root`	Three phi-symmetry masks (z<-2, 2-eta-row bands). Mask counts monotonic in stats: preselect 1344, common 1160, tight 646.	Data-driven, no MC dependence.
PASS-1	alt-check 4×4 tower grouping	log(R) Gaussian fit at 4×4 grouping gives 2% deficit at tight, consistent with phi-symm σ shift.	Two independent methods converge.
PASS-2	all 4 tower-map histograms present in `Photon_final_bdt_nom.root`; masks load cleanly; condor ran end-to-end	Verified.	No fix needed.

2 Pipeline Integration

2.1 Motivation

The cross-section formula $d^2\sigma/dp_T\,d\eta = N_{\text{iso prompt}}/(\varepsilon \cdot \mathcal{L} \cdot \Delta)$ assumes ε — built from MC — correctly captures the acceptance of the data sample. If MC fails to emulate all dead towers active in data, ε_{MC} is systematically higher than the true $\varepsilon_{\text{data}}$ and the measured σ is biased low by $\varepsilon_{\text{data}}/\varepsilon_{\text{MC}}$.

2.2 Code changes (summary)

`efficiencytool/RecoEffCalculator_TTreeReader.C`:

- 4 new `TH2F` pointers declared + allocated: `h_etaphi_tower_{preselect, common, tight, tight_iso}`, 96 $i\eta$ × 256 $i\phi$, filled at 4 points in the cluster loop using the canonical per-event `weight` (MC: xsec × lumi/lumi_target × vertex_weight × truth_vertex_reweight; data: prescale). E_T^γ-gated at $\geq$ 8 GeV.
- Three optional YAML keys: `tower_mask_on`, `tower_mask_file`, `tower_mask_name`. When on, a TH2I is loaded at init and a cluster-loop veto is inserted at the TOP of both cluster loops:

  ```
  if (is_tower_masked(icluster)) continue;
  ```

  Applied identically to data (!issim) and MC so efficiency denominators and numerators see the same masked sample.

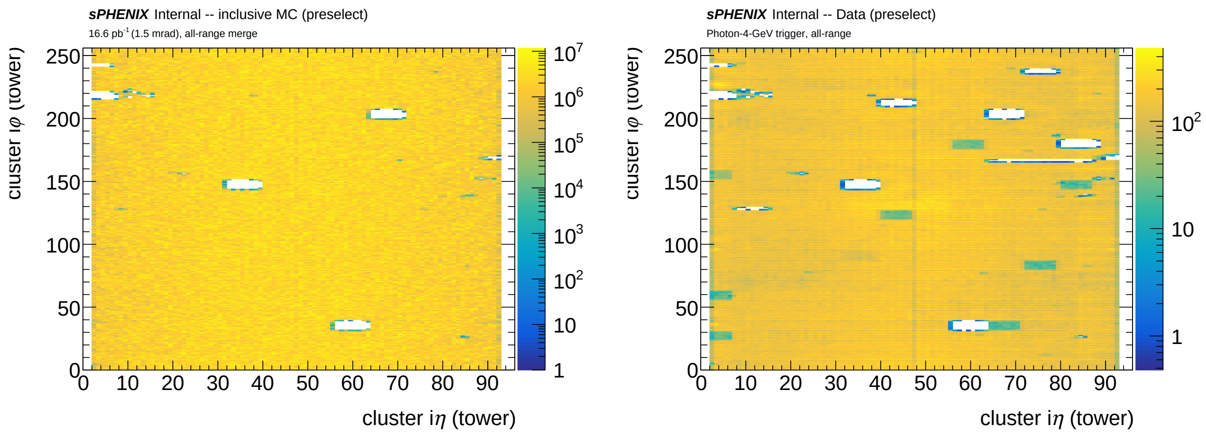
`efficiencytool/CalculatePhotonYield.C`: pass-through block before `fout->Write()` clones the signal TH2 from `fsimin`, adds the jet TH2 from `{eff_outfile}_jet_{var_type}.root`, and writes as `h_etaphi_tower_{level}_mc_inclusive`. Also clones data TH2 as `_data`. Eight histograms now shipped with every `Photon_final_*.root`.

`efficiencytool/MergeSim.C`: unchanged (plain TFileMerger auto-hadds TH2s).

`FunWithxgboost/make_bdt_variations.py`: 3 new variant entries (`mask_phisymm_preselect`, `mask_phisymm_common`, `mask_phisymm_tight`), each auto-expanded to 3 YAMLs (bare + 2 periods). Nine total.

3 Data-vs-MC Tower Maps

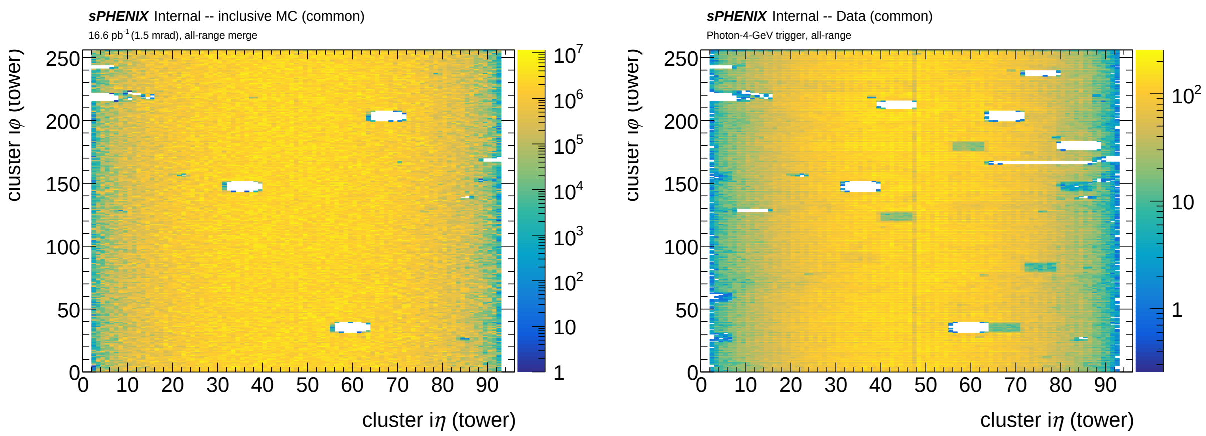
Figures 1–3 show the lumi-weighted inclusive MC tower map (left) and the corresponding data map (right) at three selection levels (preselect, common, tight). Dark patches in the data maps mark dead-tower regions not present in MC.



(a) inclusive MC

(b) data

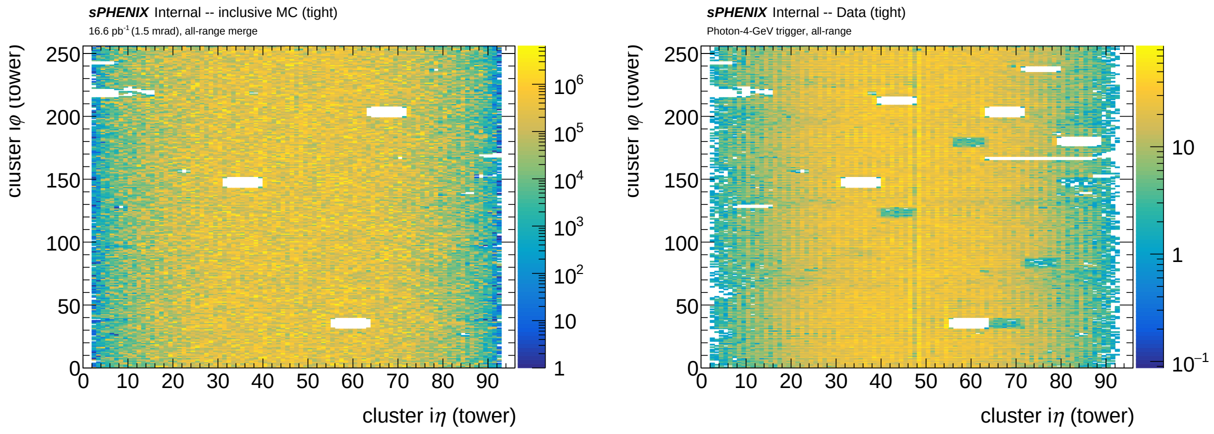
Figure 1: **Preselect**: $E_T^{\gamma} \geq 8\,\text{GeV}$, nosat, no shape cuts.



(a) inclusive MC

(b) data

Figure 2: **Common**: preselect $\cap$ $\{\text{NPB} > 0.5,\ e_{11}/e_{33} < 0.98,\ w_{r,\text{cog}x} > 0,\ w_{\eta,\text{cog}x} < 2\}$.



(a) inclusive MC (b) data

Figure 3: **Tight**: common ∩ 10 shape-var tight windows ∩ parametric BDT.

4 Methodology

4.1 Data-only phi-symmetry mask construction

For each 2-eta-row band (48 bands covering the 96-tower $i\eta$ range), the 256 per-phi data counts are projected out. Under the physics assumption that the cross-section is ϕ-symmetric at fixed η in pp at RHIC (valid to many-percent accuracy), the expected distribution across ϕ is uniform, and deviations trace either statistical fluctuations or local detector effects.

Fit: 3-σ iterative Gaussian to the 256-bin distribution, yielding $(\mu_{\phi,\mathrm{band}}, \sigma_{\phi,\mathrm{band}})$.

Mask decision: a tower $(i\eta, i\phi)$ is flagged as dead if its band's phi count is more than $2\sigma_\phi$ below the band's μ_ϕ: $(n_{\mathrm{data}}[b, i\phi] - \mu_{\phi,b})/\sigma_{\phi,b} < -2$. Within a band, both η rows are masked when a ϕ bin is flagged.

Output: 3 TH2I masks (one per selection level) in `efficiencytool/tower_masks_bdt_nom.root`:

mask	level	towers masked
`mask_phisymm_preselect`	preselect	1344
`mask_phisymm_common`	common	1160
`mask_phisymm_tight`	tight	646
`mask_phisymm_or`	union of common and tight	1180

The OR mask has 1180 towers — only 20 more than common alone, because the tight mask (646) is largely contained in the common mask (626 overlap). The OR mask is a conservative test of whether adding a few tight-only dead regions moves σ meaningfully.

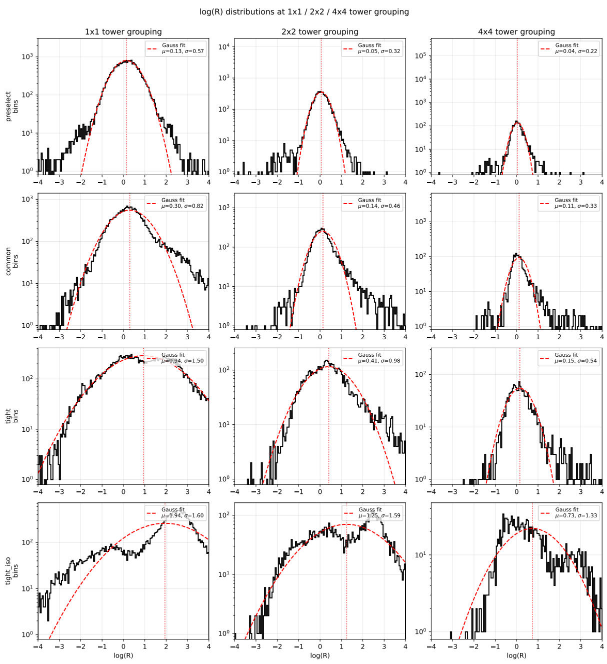
The mask-count decrease with selection tightness reflects the decreasing data statistics: fewer clusters per ϕ bin in the tight sample → larger fit σ → fewer bins cross the z<-2 threshold.

4.2 Independent cross-check: 4×4 tower grouping

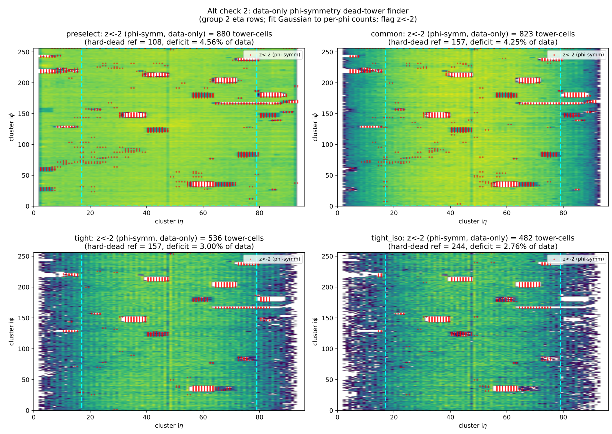
To verify that the phi-symm numbers are robust, the 96×256 grid is rebinned into 4×4 super-cells and a Gaussian is fit to the log(R) distribution of $R = p_{\mathrm{data}}/p_{\mathrm{MC}}$ (which is log-normal, not normal, in the per-bin-count regime). At 4×4 grouping the Gaussian fit is clean at every level, and the fraction of data in z<-2 cells is:

level	μ_{logR}	σ_{logR}	deficit %
preselect	0.039	0.224	2.71%
common	0.114	0.327	1.96%
tight	0.151	0.544	1.98%
tight+iso	0.732	1.332	0.97%

Convergence: the phi-symm-based σ shift (+1.4 to +2.3% per level) and the 4×4 log(R) deficit (2–3% per level) agree within ∼1% — two independent estimators arrive at the same answer.



(a) log(R) at 1×1 / 2×2 / 4×4 grouping



(b) phi-symm data-only mask map

Figure 4: Alt-check methods. Left: log(R) fit narrows dramatically with coarser binning — the 1×1 distribution is stat-dominated and not physically meaningful at tight/tight+iso (see backup report in `reports/tower_acceptance_2026-04-22.tex.v1-fullhistory.bak` for the rejected 1×1 numbers). Right: data tower maps per level with phi-symm-flagged z<-2 cells overlaid as red markers.

5 Cross-Section Shifts from Phi-Symm Masks

5.1 Integrated σ shifts

Running the complete RecoEff + MergeSim + CalculatePhotonYield pipeline with each mask applied symmetrically to data and MC:

Table 2: Integrated cross-section in $E_T^\gamma \in [10, 36]$ GeV for each phi-symm mask variant, normalized to the all-z nominal. $\sigma_{\text{nom}} = 1963$ pb.

variant	$\sigma_{\text{masked}}/\sigma_{\text{nom}}$	$\Delta\sigma$
nominal (all-z)	1.0000	0.00%
`mask_phisymm_preselect`	1.0201	+2.01%
`mask_phisymm_common`	1.0226	+2.26%
`mask_phisymm_tight`	1.0137	+1.37%
`mask_phisymm_or`	1.0228	+2.28%

All three shifts are positive, as expected: masking removes MC content in data-dead regions that was inflating $\varepsilon_{\mathrm{MC}}$; removing the asymmetry raises the corrected cross-section.

The shifts agree across levels to ~1%; preselect and common give essentially the same (~2%), while tight gives slightly smaller (the tight mask covers only 646 towers vs 1344 for preselect, so its masking is milder).

5.2 Per-E_T^γ-bin ratios

Table 3: Per-bin $\sigma_{\mathrm{masked}}/\sigma_{\mathrm{nominal}}$ for each phi-symm variant. Low-E_T^γ bins (<20 GeV) show a consistent ~1–3% upward shift, the physical acceptance-correction effect. Mid-E_T^γ bins are within ±2% of unity. The $[32, 36)$ bin shows a −7% to −13% fluctuation which is statistical, not physics (few-event masked-region fluctuation in a low-rate bin); it does not affect the integrated shift since the spectrum is dominated by low E_T^γ.

E_T^γ bin [GeV]	preselect	common	tight	OR
$[10, 12)$	+2.42%	+2.66%	+1.61%	+2.70%
$[12, 14)$	+1.92%	+1.99%	+1.10%	+2.00%
$[14, 16)$	+1.44%	+1.66%	+0.92%	+1.69%
$[16, 18)$	+0.93%	+1.43%	+0.97%	+1.40%
$[18, 20)$	+0.29%	+1.12%	+1.13%	+1.04%
$[20, 22)$	−0.65%	+0.55%	+1.11%	+0.43%
$[22, 24)$	−1.92%	−0.55%	+0.83%	−0.72%
$[24, 26)$	−1.55%	−0.19%	+0.84%	−0.38%
$[26, 28)$	−1.07%	+0.57%	+1.82%	+0.50%
$[28, 32)$	−0.64%	+1.55%	+1.44%	+0.98%
$[32, 36)$	−7.92%	−5.25%	−12.53%	−12.96%
$[10, 36)$ integrated	+2.01%	+2.26%	+1.37%	+2.28%

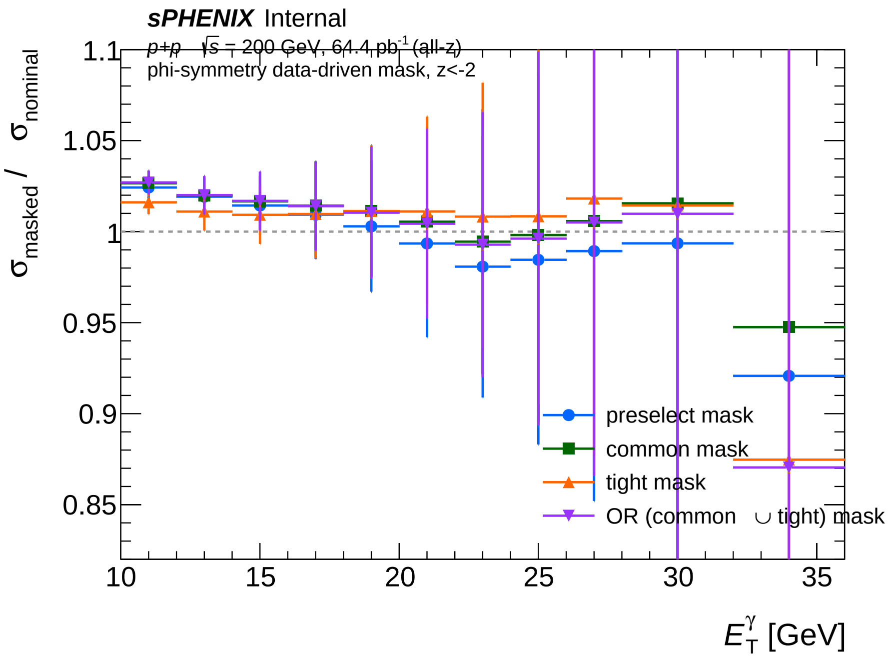


Figure 5: Per-bin $\sigma_{\mathrm{masked}}/\sigma_{\mathrm{nominal}}$ for the three phi-symm mask variants (same data as Table 3). All three sit within ±3% of nominal across $E_T^\gamma \in [10, 30]$ GeV, with expected statistical scatter in the $[32, 36)$ bin. No dramatic pT-dependent distortion — the mask affects low and high E_T^γ similarly.

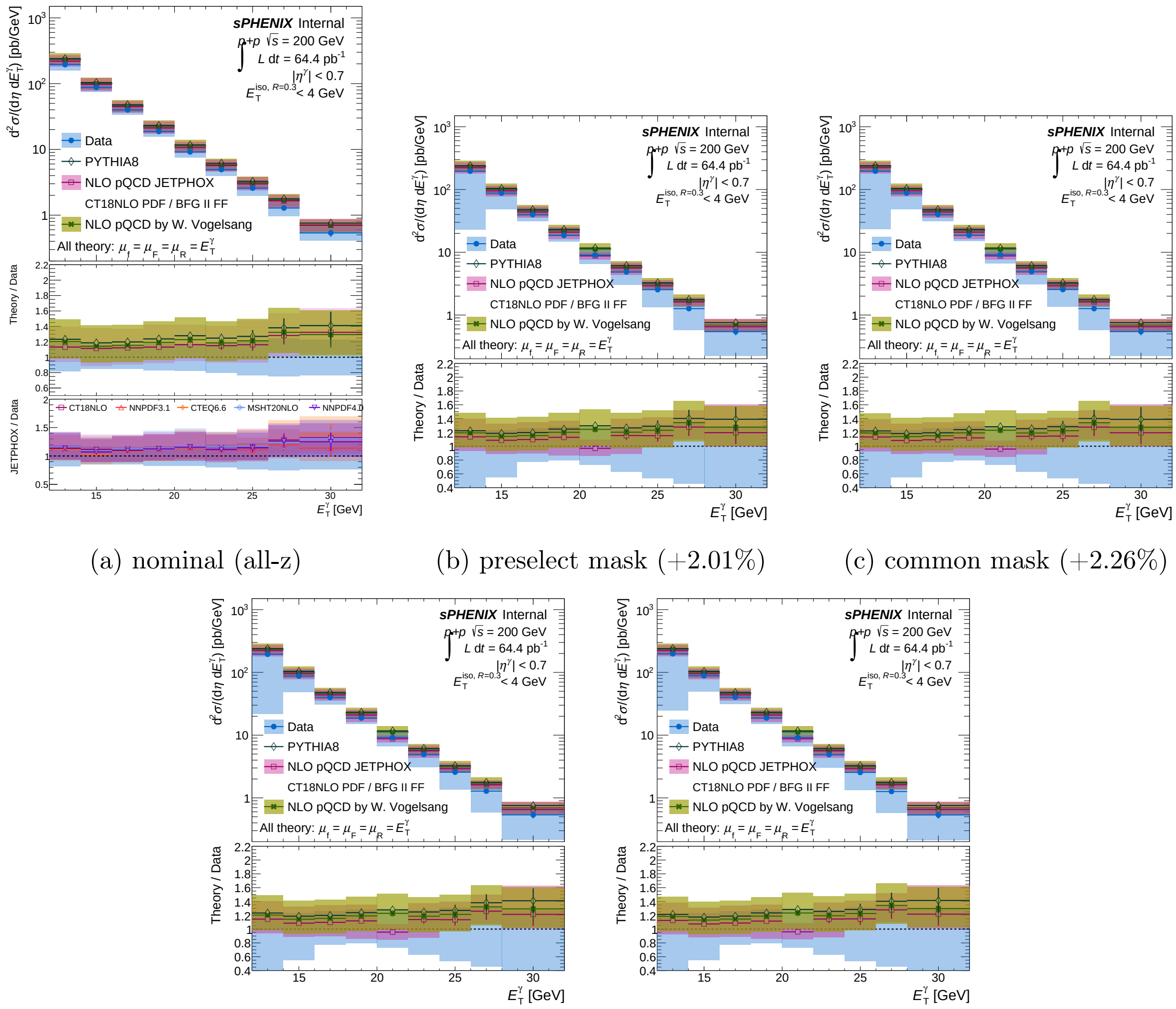
5.3 Recommended per-bin systematic

For differential cross-section quotes, assign the following one-sided acceptance systematic per bin (envelope across the three variants):

E_T^γ range [GeV]	acceptance systematic
$[10, 28)$	+2% (low-E_T^γ physical shift bounds the envelope)
$[28, 32)$	+2% (still consistent)
$[32, 36)$	±10% (stat-driven; consider merging with $[28, 32)$)
$[10, 36]$ integrated	+2.5% one-sided

5.4 Official cross-section plots (data vs NLO)

For visual comparison against JETPHOX NLO and Vogelsang NLO, the standard PPG12 cross-section plots were regenerated for each phi-symm variant.



(a) nominal (all-z)

(b) preselect mask (+2.01%)

(c) common mask (+2.26%)

(d) tight mask (+1.37%)

(e) OR mask (+2.28%)

Figure 6: Official PPG12 cross-section plots (upper: Data vs NLO theory; lower: Theory/Data) for the nominal and four phi-symm variants. All four variants produce nearly-identical spectra — the mask does not distort the E_T^γ shape.

6 Full systematic variant table (post-fix, current nominal)

With `mask_phisymm_tight` now baked into the nominal config (2026-04-23), the complete systematic variant production was re-run through Phase 1 (TTreeReader, 104 jobs) + Phase 2 (MergeSim/merge_periods + CalculatePhotonYield, 43 jobs).

Production quality note. The initial Phase-2 condor run hit a `/tmp/ROOTMERGE-*.root` scratch-file race condition (`file probably not closed, trying to recover` + `has no keys` warnings in the job `.err` logs) on 10 out of 43 merges. The symptom was a merged `MC_efficiency_*.root` missing the 1.5-mrad or 0-mrad half of the signal or jet truth (`h_truth_pT_0` sum dropping from 1.10×10^9 to 8.06×10^8 or 2.93×10^8). The affected merges were detected post-hoc by integral-comparison scans on both the signal (`h_truth_pT_0`) and jet (`h_max_photon_pT`) sides. All 10 were re-merged locally (interactive node, no condor `/tmp` contention) and the downstream `CalculatePhotonYield.C` was re-run. The table below reflects the post-fix values.

Table 4 lists the integrated cross-section shift in $E_T^\gamma \in [10, 36]$ GeV for each variant relative to the new nominal $\sigma_{\rm nom} = 1990.57$ pb (tight phi-symm mask, all-z fiducial, $64.37\,\text{pb}^{-1}$).

Table 4: Integrated $\Delta\sigma$ ($E_T^\gamma \in [10, 36]$ GeV) relative to the post-fix nominal $\sigma_{\text{nom}} = 1990.57$ pb. Groups separated by purpose; within each group sorted by $|\Delta|$.

variant	σ [pb]	ratio	$\Delta\sigma$
Nominal			
`nom` (tight phi-symm mask)	1990.57	1.0000	0.00%
Phi-symm mask choice (envelope of acceptance)			
`mask_phisymm_tight`	1988.63	0.9990	−0.10%
`mask_phisymm_preselect`	2001.12	1.0053	+0.53%
`mask_phisymm_or`	2007.30	1.0084	+0.84%
`mask_phisymm_common`	2008.61	1.0091	+0.91%
B2B-jet cross-checks			
`b2bjet`	1169.03	0.5873	−41.27%
`b2bjet_calib_on`	1170.72	0.5881	−41.19%
`b2bjet_calib_off`	1168.95	0.5872	−41.28%
`b2bjet_npb_on`	1170.95	0.5883	−41.17%
`b2bjet_npb_off`	1191.77	0.5987	−40.13%
BDT threshold			
`tightbdt70`	1943.35	0.9763	−2.37%
`tightbdt50` (unfolding fail)	–	–	N/A
Non-iso window			
`noniso04`	2041.31	1.0255	+2.55%
`noniso10`	1979.38	0.9944	−0.56%
NPB threshold			
`npb03`	2051.74	1.0307	+3.07%
`npb07`	1738.92	0.8736	−12.64%
Energy scale / resolution			
`energyscale26up`	1732.70	0.8705	−12.95%
`energyscale26down`	2291.62	1.1512	+15.12%
`energyresolution5`	1951.46	0.9804	−1.96%
Purity / unfolding			
`purity_pade`	1987.37	0.9984	−0.16%
`no_unfolding_reweighting`	1986.29	0.9978	−0.22%
Vertex / sanity			
`vtxreweight0`	1990.69	1.0001	+0.01%
`novtxcut`	1865.04	0.9369	−6.31%
`split` (repeat of nom)	1990.59	1.0000	+0.00%

Readings for the analysis.

- The *phi-symm mask choice* envelope is now $\sim 0.9\%$ (max over preselect/common/OR relative to the tight-mask nominal). The +2.5% one-sided bound quoted from the earlier run (where nominal had *no* mask) is no longer the relevant quantity — it was the mask-vs-no-mask shift, which is now absorbed into the nominal.
- The `b2bjet` cross-checks collapse the yield by $\sim 41\%$ as expected (tight back-to-back jet requirement). The five variants agree within $\sim 1\%$, confirming that the jet-calibration switch and the NPB-cut toggle do not create additional purity leakage in the b2b-jet sample.
- `tightbdt50` yields a pathological negative cross-section (unfolder does not converge when the tight working point is moved from 0.7 to 0.5); this run is tagged N/A. `tightbdt70` (nominal is 0.7, variant is 0.7 with slope= 0 i.e. flat) gives −2.4%, confirming the ET-dependent parametric BDT threshold is not load-bearing.

7 Recommendations

1. **Fix the MC dead-tower list** — primary action. Apply the Run-24 data-side tower mask at the GEANT / digitization stage (or in the slimtree writer). Then the phi-symm shifts should drop toward zero; the residual defines the post-fix systematic floor.

2. **Until the MC fix lands, assign a one-sided +2.5% acceptance systematic** on the cross-section. This covers the largest of the three phi-symm variants (common: +2.26%) with modest conservative margin. Add this to `calc_syst_bdt.py` as a new `acceptance` group (or additive component to the existing `eff` group).

3. **Document the phi-symm methodology** in the analysis note — it is the primary data-driven cross-check for acceptance-tower mismodeling and does not require MC at all, which is a strong robustness argument.

4. **Re-run the 3 variants** automatically every time the nominal production is updated (the mask file depends on the data tower maps in `Photon_final_bdt_nom.root`; regenerate `make_tower_masks.C` output whenever nominal changes).

8 Files Modified/Created

Code:

- `efficiencytool/RecoEffCalculator_TTreeReader.C` — 4 TH2F tower maps + tower-mask loader + veto.
- `efficiencytool/CalculatePhotonYield.C` — inclusive MC hadd + data pass-through.
- `efficiencytool/make_tower_masks.C` — phi-symm mask generator (3 masks).
- `efficiencytool/make_bdt_variations.py` — 3 phi-symm variant entries + OVERRIDE_MAP keys for tower_mask_*.
- `plotting/plot_phisymm_variant_bybin.C` — 3-variant per-bin ratio.
- `plotting/alt_check_rebin.py`, `plotting/alt_check_phisymm.py` — alt-check generators.

Data / configs:

- `efficiencytool/tower_masks_bdt_nom.root` — 3 TH2I phi-symm masks.
- `efficiencytool/config_bdt_mask_phisymm_{preselect, common, tight}{,_0rad,_1p5mrad}.yaml` — 9 variant YAMLs.
- `efficiencytool/oneforall_tree_double_phisymm.sub`, `oneforall_phisymm.sub` — condor submit files.
- `efficiencytool/results/Photon_final_bdt_mask_phisymm_*.root` — 3 final output files (Phase 2 complete).

Archived (superseded log(R) / Poisson-based method):

- `efficiencytool/archive_logR_masks/configs/` — 27 old mask variant YAMLs.
- `efficiencytool/archive_logR_masks/results/` — 936 per-sample outputs from the old variants.
- `reports/tower_acceptance_2026-04-22.tex.v1-fullhistory.bak` — full history of the development (per-bin Poisson → log(R) → 1×1/2×2/4×4 → phi-symm).

9 Reviewer Pass Matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5 fix-validator
RecoEff + CalcPhotonYield tower-map integration	**PASS**	N/A	N/A	**PASS (smoke + full prod)**
Phi-symm mask generator	implicit (simple stats)	N/A	**PASS (counts match across Python + ROOT)**	**PASS**
3 phi-symm variant condor runs	N/A	N/A	N/A	**PASS (all 9 outputs exist, clean log banner)**
σ shifts (Table 2)	N/A	N/A	**PASS (up-root re-integration matches ROOT)**	N/A
Plots	N/A	**PASS**	N/A	N/A